

Robotics

Omkar Tasgaonkar

otasgaonkar@ucla.edu

This is a set of notes covering interesting concepts I find while studying the Modern Robotics textbook. I will try to bring analytic theory whenever applicable.

Textbooks Used to Develop these Notes:

1. *Modern Robotics* - Kevin Lynch, Frank Park
2. *Calculus on Manifolds* - Michael Spivak
3. *Mathematical Analysis* - Tom Apostol

Other Textbooks Referenced:

1. *An Introduction to Manifolds* - Loring Tu

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NOTE: Any sections/subsections marked with an asterisk are extra sections and have been added to this document as extensions of introduced theory.

1 Constrained Configuration Spaces

1.1 Basic Multivariable Analysis

Note: The following notes may not be entirely rigorous! I have attempted to write them logically to the best of my abilities so that they follow a structured narrative.

We begin with the exploration of multivariate analysis. When we deal with a multivariate function in $\mathbb{R}$, we deal with some map of the form $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$. We can define this more formally via vector operations as the following:

$$f \left(\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \right) = \begin{bmatrix} f_1(x_1, x_2, \dots, x_n) \\ \vdots \\ f_m(x_1, x_2, \dots, x_n) \end{bmatrix} \quad (1)$$

such that $\forall j \in [1, m] : f_j : \mathbb{R}^n \rightarrow \mathbb{R}$. Our original definition of the derivative in $\mathbb{R}$ was:

$$f'(x) := \lim_{z \rightarrow x} \frac{f(z) - f(x)}{z - x}$$

or equivalently

$$\lim_{z \rightarrow x} \frac{f(z) - f(x) - f'(x)(z - x)}{z - x} = 0$$

Using the substitution $h = z - x$, we also get the equivalent definition:

$$\lim_{h \rightarrow 0} \frac{f(x + h) - f(x) - f'(x)h}{h} = 0$$

In which case, our $\epsilon - \delta$ definition would be:

$$\forall \epsilon > 0, \exists \delta > 0, \forall h \in B(0, \delta) : \frac{|f(x + h) - f(x) - f'(x)h|}{|h|} < \epsilon$$

However if we try to plainly substitute the definition for a function $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ into the first definition, we get "division of two vectors" which is not a defined operation for vector spaces (they are only abelian groups under addition).

We consider the following lemma in order to understand how to extend the derivative to the multivariate case:

Lemma 1.1. $f'(x)$ exists $\iff \left\{ \frac{1}{\delta} \sup_{z \in B(x, \delta) \setminus \{x\}} |f(z) - f(x) - f'(x)(z - x)| \right\}_{\delta > 0} \rightarrow 0$

Proof. $\implies$: If $f'(x)$ exists, then:

$$\begin{aligned} \forall \epsilon > 0, \exists \delta_0 > 0, \forall z \in B(x, \delta_0) \setminus \{x\} : \left| \frac{f(z) - f(x)}{z - x} - f'(x) \right| < \epsilon \\ \implies \forall \epsilon > 0, \exists \delta_0 > 0, \forall \delta \leq \delta_0, \forall z \in B(x, \delta) \setminus \{x\} : \left| \frac{f(z) - f(x)}{z - x} - f'(x) \right| < \epsilon \\ \implies |f(z) - f(x) - f'(x)(z - x)| < \epsilon \cdot |z - x| < \epsilon \cdot \delta \end{aligned}$$

Because this is true for all z in the punctured δ -ball around x , we have an upper bound, so we take the supremum:

$$\implies \sup_{z \in B(x, \delta) \setminus \{x\}} |f(z) - f(x) - f'(x)(z - x)| < \epsilon \cdot \delta$$

And so we rearrange the terms to get a δ -net that converges such that:

$$\forall \epsilon > 0, \exists \delta_0 > 0, \forall \delta \leq \delta_0 : \frac{1}{\delta} \sup_{z \in B(x, \delta) \setminus \{x\}} |f(z) - f(x) - f'(x)(z - x)| < \epsilon$$

Thus proving the forward direction.

$\Leftarrow$: Now for the converse, consider:

$$\forall \epsilon > 0, \exists \delta_0 > 0, \forall \delta \leq \delta_0 : \sup_{z \in B(x, \delta) \setminus \{x\}} |f(z) - f(x) - f'(x)(z - x)| < \epsilon \cdot \delta$$

Now for any arbitrary δ -ball around x where $\delta \leq \delta_0$, we have that:

$$\begin{aligned} & \sup_{z \in B(x, \delta) \setminus \{x\}} |f(z) - f(x) - f'(x)(z - x)| < \epsilon \cdot \delta \\ \implies & \forall z \in B(x, \delta) \setminus \{x\} : \frac{1}{\delta} |f(z) - f(x) - f'(x)(z - x)| < \epsilon \\ \implies & \forall z \in B(x, \delta_0) \setminus B(x, \delta) : \frac{|f(z) - f(x) - f'(x)(z - x)|}{|z - x|} < \frac{1}{\delta} |f(z) - f(x) - f'(x)(z - x)| < \epsilon \end{aligned}$$

Since δ was arbitrary as long as it was less than or equal to δ_0 , we observe:

$$\begin{aligned} & \forall \epsilon > 0, \exists \delta_0 > 0, \forall \delta \leq \delta_0, \forall z \in B(x, \delta_0) \setminus B(x, \delta) : \left| \frac{f(z) - f(x)}{z - x} - f'(x) \right| < \epsilon \\ \iff & \forall \epsilon > 0, \exists \delta_0 > 0, \forall z \in \bigcup_{\delta \in (0, \delta_0]} (B(x, \delta_0) \setminus B(x, \delta)) : \left| \frac{f(z) - f(x)}{z - x} - f'(x) \right| < \epsilon \\ \iff & \forall \epsilon > 0, \exists \delta_0 > 0, \forall z \in B(x, \delta_0) \setminus \bigcap_{\delta \in (0, \delta_0]} B(x, \delta) : \left| \frac{f(z) - f(x)}{z - x} - f'(x) \right| < \epsilon \end{aligned}$$

Now because $\mathbb{R}$ is a T_1 space, we see that the intersection in the previous statement is equivalent to $\{x\}$, so we get:

$$\forall \epsilon > 0, \exists \delta_0 > 0, \forall z \in B(x, \delta_0) \setminus \{x\} : \left| \frac{f(z) - f(x)}{z - x} - f'(x) \right| < \epsilon$$

□

We can define the function:

$$r(z) = \frac{f(z) - f(x) - f'(x)(z - x)}{z - x}$$

or $r(h)$ if we say $z = x + h$. Then taking limit as $z \rightarrow x$ (or $h \rightarrow 0$), we have that $r(z) \rightarrow 0$. Then the idea of the linear approximation is that:

$$f(z) = f(x) + f'(x)(z - x) + r(z)(z - x)$$

and the lemma above tells us that the worst remainder (similar to oscillation, which we called the "worst case net") converges to 0.

Definition 1.1. Given some $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ in the form of equation (1), then for some $i \in [1, m], j \in [1, n]$:

$$\frac{\partial f_i(x)}{\partial e_j} := \lim_{h \rightarrow 0} \frac{f_i(x + e_j h) - f_i(x)}{h}$$

where $\{e_1, \dots, e_n\}$ is the canonical basis of $\mathbb{R}^n$. This is the definition of the partial derivative.

Since each $f_i : \mathbb{R}^n \rightarrow \mathbb{R}$, we see that the partial derivative is scalar valued. We now show that it admits the linear approximation property. We can rewrite our limit definition as:

$$\lim_{h \rightarrow 0} \left(\frac{f_i(x + e_j h) - f_i(x)}{h} - \frac{\partial f_i(x)}{\partial e_j} \right) = \lim_{h \rightarrow 0} \frac{f_i(x + e_j h) - f_i(x) - \left(\frac{\partial f_i(x)}{\partial e_j} \right) h}{h} = 0$$

So with an argument similar to Lemma 1.1, we see that $f_i(z) \approx f_i(x) + \left(\frac{\partial f_i}{\partial e_j} \right) h$. Now observe that:

$$f(z) := \begin{bmatrix} f_1(z) \\ \vdots \\ f_m(z) \end{bmatrix}$$

Lemma 1.2. For all $i \in [1, m]$ and some $x \in \mathbb{R}^n$ and an e_j :

$$\frac{\partial f_i(x)}{\partial e_j} \text{ exist}$$

if and only if:

$$\frac{\partial f(x)}{\partial e_j} \text{ exists}$$

Proof. $\implies$: We claim that the partial derivative of f takes the following form:

$$\frac{\partial f(x)}{\partial e_j} = \begin{bmatrix} \frac{\partial f_1(x)}{\partial e_j} \\ \vdots \\ \frac{\partial f_m(x)}{\partial e_j} \end{bmatrix}$$

We have some δ_i given by the existence of each partial derivative, so we define $\delta_0 := \min\{\delta_i : 1 \leq i \leq n\}$. Then for all $h \in B(0, \delta_0)$:

$$\frac{\|f(x + e_j h) - f(x) - \left(\frac{\partial f(x)}{\partial e_j} \right) h\|}{|h|} \leq \frac{1}{|h|} \sum_{i=1}^m \left| f_i(x + e_j h) - f_i(x) - \left(\frac{\partial f_i(x)}{\partial e_j} \right) h \right| < m\epsilon$$

where we used the inequality that the $\|v\| \leq \sum |v_i|$.

$\Leftarrow$: We claim that:

$$\frac{\partial f_i(x)}{\partial e_j} = e_i \cdot \frac{\partial f(x)}{\partial e_j}$$

We now use the inequality $|v_i| \leq \|v\|$ and find that $\forall i \in [1, m]$:

$$\frac{|f_i(x + e_j h) - f_i(x) - (e_i \cdot \frac{\partial f(x)}{\partial e_j}) h|}{|h|} = \frac{|e_i \cdot f(x + e_j h) - e_i \cdot f_i(x) - (e_i \cdot \frac{\partial f(x)}{\partial e_j}) h|}{|h|}$$

$$= \frac{|e_i \cdot (f(x + e_j h) - f(x) - \frac{\partial f(x)}{\partial e_j} h)|}{|h|} \leq \frac{\|f(x + e_j h) - f(x) - (\frac{\partial f(x)}{\partial e_j} h)\|}{|h|} < \epsilon$$

so the same δ suffices. □

Lemma 1.3. *The partial derivative of $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ exists for some e_j if and only if there exists some $r(h) : \mathbb{R} \rightarrow \mathcal{M}_{m \times 1}(\mathbb{R})$ such that:*

$$f(x + h e_j) = f(x) + \frac{\partial f(x)}{\partial e_j} h + r(h)h \wedge \lim_{h \rightarrow 0} r(h) = 0$$

Proof. $\implies$: By the previous Lemma 1.2 and the results of Lemma 1.1, we can say that there exists some $r_i(h)$ for each f_i , such that:

$$r_i(h) = \frac{f_i(x + h e_j) - f_i(x) - \frac{\partial f_i(x)}{\partial e_j} h}{h}$$

Then, we can define $r(h)$ as:

$$r(h) = \begin{bmatrix} r_1(h) \\ \vdots \\ r_m(h) \end{bmatrix}$$

which tends to 0 by definition, or by the fact that $\|v\| \leq \sum |v_i|$.

$\Leftarrow$: We can rewrite the expression as:

$$r(h) = \frac{f(x + h e_j) - f(x) - \frac{\partial f(x)}{\partial e_j} h}{h}$$

so $\|r(h)\| \rightarrow 0$ implies that the quotient also tends to zero. □

Notice that the proofs did not use any properties of the basis vector e_j itself, so we can extend the definition:

Definition 1.2. *Given some $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ in the form of equation (1), then for some $v \in \mathbb{R}^n$ and $i \in [1, m], j \in [1, n]$:*

$$\frac{\partial f_i(x)}{\partial v} := \lim_{h \rightarrow 0} \frac{f_i(x + h v) - f_i(x)}{h}$$

and similarly for f . Such a derivative is called the Gateaux derivative.

We can extend the Gateaux derivative to obtain the Frechet derivative. The Gateaux derivative limited itself to a specific direction of the vector v (and then used h as the variable for the limit).

Definition 1.3. *The Frechet derivative of a function f at $x \in \mathbb{R}^n$ is:*

$$Df(x) : \lim_{v \rightarrow 0} \frac{\|f(x + v) - f(x) - (Df(x))(v)\|}{\|v\|} = 0$$

or equivalently:

$$\forall \epsilon > 0, \exists \delta > 0, \forall v \in B(0, \delta) : \frac{\|f(x + v) - f(x) - (Df(x))(v)\|}{\|v\|} < \epsilon$$

where the norm induces a metric so that $v \in B(0, \delta)$ (in this case 0 is the 0 vector), is the same as saying $\|v\| \in B(0, \delta)$ or equivalently $\|v\| < \delta$.

The form of the limit is similar to the idea of linear approximation. Since $v \in \mathbb{R}^n$, then we must have that $Df(x) \in \mathcal{M}_{m \times n}(\mathbb{R})$. If the function is Frechet differentiable, then $Df(x)$ is the Jacobian:

$$Df(x) = \begin{bmatrix} \frac{\partial f_1(x)}{\partial e_1} & \dots & \frac{\partial f_1(x)}{\partial e_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m(x)}{\partial e_1} & \dots & \frac{\partial f_m(x)}{\partial e_n} \end{bmatrix} = \begin{bmatrix} \frac{\partial f(x)}{\partial e_1} & \dots & \frac{\partial f(x)}{\partial e_n} \end{bmatrix}$$

Proof. If we chose the family of vectors $\{he_j : h \in \mathbb{R} \wedge j \in [1, n]\}$, then differentiability tells us:

$$\forall \epsilon > 0, \exists \delta > 0, \forall h \in B(0, \delta) : \frac{\|f(x + he_j) - f(x) - h(Df(x))e_j\|}{|h|} < \epsilon$$

telling us that:

$$\frac{\partial f(x)}{\partial e_j} = Df(x)e_j$$

which is the j -th column of the Jacobian matrix. □

Notice that Lemma 1.2 was a statement on equivalence; however, the converse is not necessarily true for the Jacobian matrix. The total differentiability of f at x gives us the existence of partial derivatives, but not necessarily the converse. In order to prove this, we require the following concept:

1.1.1 MVT in the Multivariable Case

Theorem 1.1. *Given a function $f : \mathbb{R} \rightarrow \mathbb{R}$ that is continuous on the closed interval $[a, b]$ and differentiable on (a, b) , then:*

$$\exists z \in (a, b) : f'(z) = \frac{f(b) - f(a)}{b - a}$$

Proof. Since f is continuous on a compact interval, it achieves the extrema by EVT. Now consider the function:

$$h(x) = \frac{f(b) - f(a)}{b - a}(x - a) - f(x) + f(a)$$

which at both $x = a$ and $x = b$, we have that $h(x) = 0$, so by Rolle's Theorem, there exists some extrema within the interval, so that:

$$\exists z \in (a, b) : h'(z) = 0 \implies f'(z) = \frac{f(b) - f(a)}{b - a}$$

□

If we try to apply this proof to a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$, then in order to apply Theorem 1.1, we require some compact interval given f is continuous.

Now, the following lemma will be useful in exploring MVT:

Lemma 1.4. *Any open, convex subset $U \subseteq \mathbb{R}^n$ is path connected.*

Proof. Openness does not play an important role here. We can simply take the continuous function:

$$f(\alpha) = \alpha x + (1 - \alpha)y$$

for some fixed $x, y \in U$. By convexity, we know $\text{Ran}(f) \subseteq U$. □

Lemma 1.5. *An open ball $B(x, r)$ is convex.*

Proof. Take $y, z \in B(x, r)$, then:

$$|\alpha y + (1 - \alpha)z - x| = |\alpha(y - x) + (1 - \alpha)(z - x)| \leq \alpha r + (1 - \alpha)r = r$$

□

Theorem 1.2. *Given $f_i : \mathbb{R}^n \rightarrow \mathbb{R}$, we have that for some open, convex subset $U \subseteq \mathbb{R}^n$ (on which f_i is differentiable) and given two points a, b :*

$$\exists z \in U : Df_i(z)(b - a) = f_i(b) - f_i(a)$$

where $Df_i(z)$ is the gradient vector at z .

Proof. Since U is convex, we can take the function for $t \in [0, 1]$:

$$h(t) = tb + (1 - t)a$$

such that $h(0) = a$ and $h(1) = b$. Then applying Theorem 1.1 to the function $f_i \circ h$ (which too is continuous on $[0, 1]$ and differentiable on $(0, 1)$), we see that:

$$\exists c \in (0, 1) : (f_i \circ h)'(c) = \frac{f_i(h(1)) - f_i(h(0))}{1 - 0} = f_i(b) - f_i(a)$$

and $(f_i \circ h)'(c) = Df_i(h(c)) \cdot h'(c) = Df_i(c)(b - a)$, where:

$$Df_i(x) = \left[\frac{\partial f_i(x)}{\partial e_1} \quad \dots \quad \frac{\partial f_i(x)}{\partial e_n} \right]$$

We used the multivariate chain rule in this proof, where $(f \circ h)'(t) = Df(h(t))Dh(t)$. In this case, we defined:

$$h(t) = \begin{bmatrix} tb_1 + (1 - t)a_1 \\ \vdots \\ tb_n + (1 - t)a_n \end{bmatrix}$$

and since each $h_i : \mathbb{R} \rightarrow \mathbb{R}$, then $Dh(t)$ is an $n \times 1$ matrix in the e_1/t direction, the form of which is given by Lemma 1.2:

$$Dh(t) = \begin{bmatrix} b_1 - a_1 \\ \vdots \\ b_n - a_n \end{bmatrix} = b - a$$

□

Corollary 1.1. *If $\exists k \in \mathbb{R}, \exists j \in [1, n] : b - a = ke_j$ and the partial derivative of f_i exists in the e_j direction, then:*

$$\exists z \in \mathbb{R}^n : \frac{\partial f_i(z)}{\partial e_j} k = f_i(b) - f_i(a) \wedge z_j \in (a, b)$$

We can say the same thing for a vector valued function $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$.

Notice this corollary does not require full differentiability of f_i (which means it does not require the gradient to be the total derivative of f at said point). We can expand MVT to the vector valued function case with an inequality.

Corollary 1.2. Given $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and an open, convex subset $U \subseteq \mathbb{R}^n$ on which f is totally differentiable and $\|Df(x)\|$ is bounded, then:

$$\forall a, b \in U : \|f(b) - f(a)\| \leq M\|b - a\|$$

such that $\exists M \in \mathbb{R}, \forall x \in U : \|Df(x)\| \leq M$.

Proof. Consider any arbitrary $v \in \mathbb{R}^n$, then:

$$\forall i \in [1, m] : \frac{\|Df_i(v)v\|}{\|v\|} \leq \frac{\|Df(v)v\|}{\|v\|} \leq M$$

Thus $\|Df_i(v)\| \leq M$, so we can apply Theorem 1.2 and see the following:

$$\|f(b) - f(a)\| \leq \sum_{i=1}^m \|f_i(b) - f_i(a)\| \leq \sum_{i=1}^m \|Df_i(z_i)\| \cdot \|b - a\| \leq mM\|b - a\|$$

This is not the estimate we were looking for, but one possible estimate. We can derive another by the following:

$$\begin{aligned} \|f(b) - f(a)\| &= \sqrt{\sum_{i=1}^m |f_i(b) - f_i(a)|^2} = \sqrt{\sum_{i=1}^m |Df_i(z_i)(b - a)|^2} \\ &\leq \sqrt{\sum_{i=1}^m C^2 \|b - a\|^2} = \sqrt{m} \cdot C \|b - a\| \end{aligned}$$

Consider once again the intermediate function $h(t)$ where $h : [0, 1] \rightarrow \mathbb{R}^n$ introduced in our proof of Theorem 1.2, then:

$$g(t) = (f(b) - f(a))^T f(h(t)) \implies g'(t) = (f(b) - f(a))^T Df(h(t))(b - a)$$

thus $g'(t) : \mathbb{R} \rightarrow \mathbb{R}$. Then define $g : [0, 1] \rightarrow \mathbb{R}$:

$$|g'(t)| \leq \|f(b) - f(a)\| \cdot \|Df(h(t))\| \cdot \|b - a\| \leq \|f(b) - f(a)\| \cdot M\|b - a\|$$

We can apply regular MVT to g , which tells us:

$$|g(1) - g(0)| = |(f(b) - f(a))^T (f(b) - f(a))| = \|f(b) - f(a)\|^2 \leq \|f(b) - f(a)\| \cdot M\|b - a\|$$

and the intended inequality follows. $\square$

Lemma 1.6. The function $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is totally differentiable at a point if:

1. The partial derivatives exist
2. The partial derivatives are continuous at that point

Proof. We want to show:

$$\forall \epsilon > 0, \exists \delta > 0, \forall v \in B(0, \delta) : \frac{\|f(x+v) - f(x) - (Df(x))v\|}{\|v\|} < \epsilon$$

The expression $(Df(x))(v)$ gives us the following vector:

$$(Df(x))(v) = \begin{bmatrix} \sum_{i=1}^n \frac{\partial f_1(x)}{\partial e_i} v_i \\ \vdots \\ \sum_{i=1}^n \frac{\partial f_m(x)}{\partial e_i} v_i \end{bmatrix}$$

and we see that:

$$\frac{\|f(x+v) - f(x) - (Df(x))(v)\|}{\|v\|} \leq \frac{1}{\|v\|} \sum_{j=1}^m \left| f_j(x+v) - f_j(x) - \sum_{i=1}^n \frac{\partial f_j(x)}{\partial e_i} v_i \right|$$

Continuity of the partial derivatives give us:

$$\forall \epsilon > 0, \exists \delta_{ji} > 0, \forall v \in B(0, \delta_{ji}) : \left| \frac{\partial f_j(x+v)}{\partial e_i} - \frac{\partial f_j(x)}{\partial e_i} \right| < \epsilon$$

We set $\delta := \min\{\delta_{ji} : 1 \leq j \leq m \wedge 1 \leq i \leq n\}$. Then given some $v \in B(0, \delta)$, we apply the idea in Corollary 1.1 and see that:

$$f_j(x+v) - f_j(x) = \sum_{i=1}^n f_j(x+V_i) - f_j(x+V_{i-1})$$

where we define $V_i := v_1 e_1 + \dots + v_i e_i$. Then, we can repeatedly apply Corollary 1.1 and get:

$$= \frac{1}{\|v\|} \sum_{j=1}^m \left| \sum_{i=1}^n \left(\frac{\partial f_j(z_i)}{\partial e_i} - \frac{\partial f_j(x)}{\partial e_i} \right) v_i \right| \leq \frac{\epsilon \cdot mn \cdot \|v\|}{\|v\|} = \epsilon \cdot mn$$

□

The MVT introduced in Theorem 1.2 fails for vector valued functions as we cannot guarantee the same z satisfies the equality for all f_i . Finally, we can expand the idea of linear approximation to the total vector valued function:

Lemma 1.7. $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is totally differentiable if and only if there exists $r(v) : \mathbb{R}^n \rightarrow \mathcal{M}_{m \times n}(\mathbb{R})$ such that:

$$f(x+v) = f(x) + Df(x)(v) + r(v)v \wedge \lim_{v \rightarrow 0} \|r(v)\| = 0$$

Proof. Observe that we use the matrix norm, which is defined as:

$$\begin{aligned} \|A\| &= \sup_{x \in \mathbb{R}^n} \frac{\|Ax\|}{\|x\|} \\ \forall x \in \mathbb{R}^n : \frac{\|Ax\|}{\|x\|} &= \sqrt{\frac{\sum_{i=1}^m (\sum_{j=1}^n A_{ij} x_j)^2}{\sum_{j=1}^n x_j^2}} \\ &\leq \sqrt{\frac{\sum_{i=1}^m (\sum_{j=1}^n A_{ij}^2 \sum_{j=1}^n x_j^2)}{\sum_{j=1}^n x_j^2}} = \sqrt{\sum_{i=1}^m \left(\sum_{j=1}^n A_{ij}^2 \right)} \end{aligned}$$

and since this is true for all $x \in \mathbb{R}^n$, we have a concrete upper bound, so:

$$\|A\| \leq \sqrt{\sum_{i=1}^m \left(\sum_{j=1}^n A_{ij}^2 \right)}$$

Now, we can use the fact that $f_i : \mathbb{R}^n \rightarrow \mathbb{R}$ and that its total derivative exists as seen by:

$$\frac{|f_i(x+v) - f_i(x) - Df_i(x)(v)|}{\|v\|} = \frac{|e_i^T (f(x+v) - f(x) - Df(x)(v))|}{\|v\|} \leq \frac{\|f(x+v) - f(x) - Df(x)(v)\|}{\|v\|}$$

Then, we say that equivalently there exists some $r_i(v) : \mathbb{R}^n \rightarrow \mathcal{M}_{1 \times n}$ such that:

$$r_i(v)v = f_i(x+v) - f_i(x) - Df_i(x)(v)$$

such that:

$$r_i(v) = \left[(f_i(x+v) - f_i(x) - Df_i(x)(v)) \frac{v_1}{\|v\|^2} \quad \cdots \quad (f_i(x+v) - f_i(x) - Df_i(x)(v)) \frac{v_n}{\|v\|^2} \right]$$

$$\implies \|r_i(v)\| = \|f_i(x+v) - f_i(x) - Df_i(x)(v)\| \sqrt{\frac{1}{\|v\|^2}} = \frac{\|f_i(x+v) - f_i(x) - Df_i(x)(v)\|}{\|v\|} \rightarrow 0$$

The converse follows from the fact that $\|r_i(v)v\| \leq \|r_i(v)\| \cdot \|v\|$. Now, we can define:

$$r(v) := \begin{bmatrix} r_1(v) \\ \vdots \\ r_m(v) \end{bmatrix}$$

for which by the inequality for matrix norm we derived above, we have that:

$$\|r(v)\| \leq \sqrt{\sum_{i=1}^m (f_i(x+v) - f_i(x) - Df_i(x)(v))^2} \cdot \frac{1}{\|v\|^2} = \frac{1}{\|v\|} \cdot \|f(x+v) - f(x) - Df(x)(v)\| \rightarrow 0$$

This is a tighter ϵ -bound than using the fact (which we can prove) that:

$$\|r(v)\| \leq \sum_{i=1}^m \|r_i(v)\| < m\epsilon$$

taking δ minimum (follows from the fact that if f is differentiable, then f_i is differentiable). The converse follows from the fact that $\|r(v)v\| \leq \|r(v)\| \cdot \|v\|$. $\square$

Now we can prove the following:

Lemma 1.8. *If $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is differentiable at x , then it is continuous at x .*

Proof. By Lemma 1.7, we have that there exists some $r(v) : \mathbb{R}^n \rightarrow \mathcal{M}_{m \times n}(\mathbb{R})$ such that:

$$f(x+v) = f(x) + Df(x)(v) + r(v)v$$

Now, let $\|r(v)\| < 1$, which gives us some δ_0 such that for all $v \in B(0, \delta_0)$:

$$\begin{aligned} \|f(x+v) - f(x)\| &= \|(Df(x) + r(v))v\| \leq \|(Df(x) + r(v))\| \cdot \|v\| \\ &\leq \|v\| \cdot (\|Df(x)\| + 1) \end{aligned}$$

So taking:

$$\delta := \min \left\{ \delta_0, \frac{\epsilon}{1 + \|Df(x)\|} \right\}$$

we see that:

$$\forall v \in B(0, \delta) : \|f(x+v) - f(x)\| \leq \|v\| \cdot (\|Df(x)\| + 1) \leq \epsilon$$

$\square$

1.2 Inverse and Implicit Function Theorems

Note: In the rest of these notes, when I say a function f is " C^1 at x_0 " (this is nonstandard), it is my shorthand for saying that f is differentiable in a neighborhood of x_0 and Df is continuous at x_0 .

The idea of the inverse function theorem stems from our idea of the linear approximation where $f(x+v) \approx f(x) + Df(x)v$. Then $v \approx Df(x)^{-1}(f(x+v) - f(x))$. Given the recurring idea of the error function where:

$$f(x+h) = f(x) + f'(x)h + r(h)h$$

then if $r(h)$ is small enough (and $f'(x)$ is invertible), we can find a local region where f is injective, thus has a local inverse, and the other properties follow.

Theorem 1.3 (Banach Fixed Point Theorem). *Given a contraction map $f : X \rightarrow X$ and (X, ϱ) a complete metric space, such that $\varrho(f(x), f(y)) \leq c\varrho(x, y)$, where $c \in [0, 1)$, then there exists a unique fixed point such that:*

$$\exists x' \in X : f(x') = x'$$

Proof. Let us choose an arbitrary point $x \in X$. Then:

$$\varrho(f(x), f(f(x))) \leq c\varrho(x, f(x))$$

Then, we see that:

$$\varrho(f^n(x), f^{n+1}(x)) \leq c^n \varrho(x, f(x))$$

So for any arbitrary $m \leq n$:

$$\begin{aligned} \varrho(f^m(x), f^n(x)) &\leq \sum_{i=0}^{n-m-1} \varrho(f^{m+i}(x), f^{m+i+1}(x)) \leq \varrho(x, f(x)) \sum_{i=0}^{n-m-1} c^{m+i} \\ &= \varrho(x, f(x)) \sum_{i=m}^{n-1} c^i = \frac{c^m(1 - c^{n-m+1})}{1 - c} \cdot \varrho(x, f(x)) \end{aligned}$$

We now show that the sequence $\{f^{(n)}(x)\}_{n \in \mathbb{N}}$ is Cauchy:

$$\frac{c^m(1 - c^{n-m+1})}{1 - c} \cdot \varrho(x, f(x)) \leq \frac{c^m}{1 - c} \cdot \varrho(x, f(x)) < \epsilon$$

So, if we choose n_0 so that:

$$\frac{c^{n_0}}{1 - c} \cdot \varrho(x, f(x)) < \epsilon$$

then for all $m, n \geq n_0$, we have that the sequence is Cauchy. Because X is complete, we have that there exists some $x' \in X$ such that:

$$f^n(x) \rightarrow x'$$

Now, f is c -Lipschitz, so by continuity, we have that the sequence $\{f^{n+1}(x)\}_{n \in \mathbb{N}}$ converges to $f(x')$. Since it is a subsequence of our initial sequence, they must converge to the same limit, thus $f(x') = x'$.

The fixed point is also guaranteed to be unique by the uniqueness of a limit in a T_2 space, which we shall prove. Let us assume there exists another fixed point $\tilde{x}$ such that $f(\tilde{x}) = \tilde{x}$. Then:

$$\varrho(f(\tilde{x}), f(x')) = \varrho(\tilde{x}, x') \leq c\varrho(\tilde{x}, x') < \varrho(\tilde{x}, x')$$

which is a contradiction. However, if we allowed $c = 1$, then we may not have a unique fixed point (the Cauchy estimate may also fail). $\square$

Theorem 1.4 (Inverse Function Theorem). *If $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is differentiable on an open ball $B(x_0, \delta_0)$ with $Df(x_0)$ continuous and invertible, then there exists some $B(x_0, \delta) \subseteq B(x_0, \delta_0)$, then:*

1. f is injective on $B(x_0, \delta)$ and an open map
2. f^{-1} is differentiable on $f(B(x_0, \delta))$
3. $D(f^{-1})$ is continuous at $f(x_0)$

Proof. Part One

We are given that $\forall x \in B(x_0, \delta) : \|Df(x) - Df(x_0)\| < \epsilon$. We want to show that:

$$\forall x, y \in B(x_0, \delta) : f(x) = f(y) \implies x = y$$

We can define $\delta := \min\{\delta_0/2, \delta_1\}$ where δ_1 is given by continuity so that $B(x_0, \delta) \subseteq B(x_0, \delta_0)$ for $\epsilon = \frac{1}{2(\|Df^{-1}(x_0)\|+1)}$, then consider the function:

$$h(x) = x + Df(x_0)^{-1}(y - f(x))$$

for some arbitrary $y \in \text{Ran}(f|_{B(x_0, \delta)})$. We show that h is a contraction map:

$$Dh(x) = I - Df^{-1}(x_0)Df(x) = Df(x_0)^{-1}(Df(x_0) - Df(x))$$

$$\|Dh(x)\| \leq \|Df(x_0)^{-1}\| \cdot \|Df(x_0) - Df(x)\| \leq \frac{1}{2}$$

and then by Corollary 1.2, we have that:

$$\forall x, y \in B(x_0, \delta) : \|h(x) - h(y)\| \leq \frac{1}{2}\|x - y\|$$

so it is a contraction map, thus any fixed points are unique, giving us that f is injective (and furthermore trivially surjective). Notice that $B(x_0, \delta)$ is not closed (thus not complete), so we cannot apply fixed point theorem. However, we do not require fixed point theorem since the fixed points are guaranteed as $f(B(x_0, \delta))$ is the image of the open ball.

Part One (Alternate Proof)

To show injectivity, it suffices to show that there exists some $c \in \mathbb{R}$ such that:

$$\forall x_1, x_2 \in B(x_0, \delta) : \|x_1 - x_2\| \leq c\|f(x_1) - f(x_2)\|$$

We consider the equivalent criterion of differentiability whereby:

$$f(x_0 + v) = f(x_0) + Df(x_0)(v) + r(v)v \wedge \|r(v)\| \rightarrow 0$$

We can define $R(v) = r(v)v$, for which we can prove the equivalent condition:

$$f(x_0 + v) = f(x_0) + Df(x_0)(v) + R(v) \wedge \frac{\|R(v)\|}{\|v\|} \rightarrow 0$$

So, we see that $DR(v) = Df(x_0 + v) - Df(x_0)$, so by continuity, we see that, taking the same ϵ as before:

$$\forall v \in B(0, \delta) : \|DR(v)\| = \|Df(x_0 + v) - Df(x_0)\| \leq \frac{1}{2\|Df(x_0)^{-1}\|}$$

Thus by Corollary 1.2, since $B(x_0, \delta)$ is open and convex:

$$\begin{aligned} \forall v, w \in B(0, \delta) : \|R(v) - R(w)\| &\leq \frac{1}{2\|Df(x_0)^{-1}\|} \|v - w\| \\ R(v) - R(w) &= f(x_0 + v) - f(x_0) - Df(x_0)(v) - f(x_0 + w) + f(x_0) + Df(x_0)(w) \\ \|R(v) - R(w)\| &= \|f(x_0 + v) - f(x_0 + w) - Df(x_0)(v - w)\| \\ \implies \|Df(x_0)(v - w)\| - \|f(x_0 + v) - f(x_0 + w)\| &\leq \frac{1}{2\|Df(x_0)^{-1}\|} \|v - w\| \end{aligned}$$

We also know that $\|x\| = \|Df(x_0) \cdot Df(x_0)^{-1}x\| \leq \|Df(x_0)^{-1}\| \cdot \|Df(x_0)x\|$, thus

$$\implies \frac{1}{2\|Df(x_0)^{-1}\|} \|v - w\| \leq \|f(x_0 + v) - f(x_0 + w)\|$$

and a simple algebraic rearrangement brings us to the desired result.

The idea of this contraction map and the alternative proof both come from our idea of linear approximation, where:

$$f(z) \approx f(x) + Df(x)(z - x)$$

Part Two

We now prove that f is an open map, so we want to show that $f(B(x_0, \delta))$ is open:

$$\forall x \in B(x, \delta), \exists \delta' > 0 : B(f(x), \delta') \subseteq f(B(x_0, \delta))$$

The consider some $\tilde{\delta}$ such that $\overline{B(x, \tilde{\delta})} \subseteq B(x_0, \delta)$. Then we claim that the ball $\epsilon\tilde{\delta}$ is the δ' we are looking for. We use the same idea of a contraction map, but this time apply fixed point theorem. The reason we use fixed point theorem is we want to show that:

$$\forall y \in B(f(x), \epsilon\tilde{\delta}), \exists z \in \overline{B(x, \tilde{\delta})} : f(z) = y$$

telling us that $B(f(x), \epsilon\tilde{\delta}) \subseteq f(\overline{B(x, \tilde{\delta})}) \subseteq f(B(x_0, \delta))$. So for a fixed $y \in B(f(x), \epsilon\tilde{\delta})$:

$$\begin{aligned} \|h(z) - x\| &\leq \|h(z) - h(x)\| + \|h(x) - x\| \leq \frac{1}{2}\|z - x\| + \|Df(x_0)^{-1}\| \cdot \|y - f(x)\| \\ &< \frac{\tilde{\delta}}{2} + \|Df(x_0)^{-1}\| \epsilon\tilde{\delta} = \tilde{\delta} \end{aligned}$$

so h which we showed is a contraction map, is a contraction map on $\overline{B(x, \tilde{\delta})}$, thus by Banach Fixed Point theorem, for each arbitrary y , there exists a fixed point (note we need the theorem because we are not immediately guaranteed the existence of some z such that $f(z) = y$).

Part Three

We now need to show the existence of $Df^{-1}(x)$ for $x \in B(x_0, \delta)$. A matrix is invertible if and only if all of its eigenvalues are nonzero. Thus:

$$\forall i \in [1, n] : Df(x)v_i = \lambda_i v_i \wedge \lambda_i \neq 0$$

We can rewrite $Df(x)$ as $Df(x_0) + (Df(x) - Df(x_0))$, so we see that:

$$\|Df(x_0)x + (Df(x) - Df(x_0))x\| = \|Df(x_0)(x + Df(x_0)^{-1}(Df(x) - Df(x_0))x)\|$$

$$\begin{aligned} &\geq \frac{1}{\|Df(x_0)^{-1}\|} (\|x\| - \|Df(x_0)^{-1}\| \cdot \|Df(x) - Df(x_0)\| \cdot \|x\|) \\ &\geq \frac{\|x\|}{\|Df(x_0)^{-1}\|} (1 - \|Df(x_0)^{-1}\| \cdot \|Df(x) - Df(x_0)\|) \geq \frac{\|x\|}{2\|Df(x_0)^{-1}\|} \end{aligned}$$

Now assume $Df(z)$ is non-invertible, which means it has some v_i with $\lambda_i = 0$, then:

$$0 \leq \frac{\|v_i\|}{2\|Df(x_0)^{-1}\|} \leq 0$$

which forces $v_i = 0$, which is a contradiction as an eigenvector cannot be the zero vector. Thus $Df(x)$ is invertible for $x \in B(x_0, \delta)$ (observe it is only true for this ball as continuity gives us a strict lower bound of 0). Otherwise, it may be that:

$$\|Df(x_0)^{-1}\| \cdot \|Df(x) - Df(x_0)\| > 1$$

in which case the argument does not hold, since we do not get a squeeze inequality. We now must show that $Df^{-1}(f(x)) = Df(x)^{-1}$, given the invertibility of the derivative (which we just proved), injectivity of the function (proved in part one), and continuity of the inverse (also proved in part one).

We use the ideas of Lemma 1.1 and prove the result for $f : \mathbb{R} \rightarrow \mathbb{R}$. Because f is injective (and in this case a bijection as well), then f^{-1} is a valid function, so combined with the continuity of f^{-1} and invertibility of the derivative at a point x , we want to show there exists some $\hat{r}(y)$ such that:

$$f^{-1}(y) = f^{-1}(f(x)) + \frac{1}{f'(x)}(y - f(x)) + \hat{r}(y - f(x))(y - f(x))$$

and

$$\lim_{y \rightarrow f(x)} \hat{r}(y) = 0$$

And we have the existence of some $r(z) : \mathbb{R} \rightarrow \mathbb{R}$ such that:

$$f(z) = f(x) + f'(x)(z - x) + r(z - x)(z - x)$$

Then by injectivity, we can define $z := f^{-1}(y)$, so that we get:

$$\begin{aligned} y - f(x) &= (f'(x) + r(f^{-1}(y) - f^{-1}(f(x))))(f^{-1}(y) - f^{-1}(f(x))) \\ \implies f^{-1}(y) - f^{-1}(f(x)) &= \frac{y - f(x)}{f'(x) + r(f^{-1}(y) - f^{-1}(f(x)))} = \frac{y - f(x)}{f'(x)} + \frac{1}{f'(x)}(y - f(x)) \end{aligned}$$

So, we can define:

$$\hat{r}(y - f(x)) = \frac{1}{f'(x) + r(f^{-1}(y) - f^{-1}(f(x)))} - \frac{1}{f'(x)}$$

so for all nets that converge to $f(x)$, by continuity, we have that the image of those nets under f^{-1} converges to x , which means:

$$f^{-1}(y) \rightarrow x \implies r(f^{-1}(y) - x) \rightarrow 0 \implies \hat{r}(y - f(x)) \rightarrow 0$$

because we have that for all nets that converge to x , the remainder converges to 0. So we have that:

$$(f^{-1})'(f(x)) = \frac{1}{f'(x)}$$

We can expand this same idea to vector valued functions and define:

$$\hat{r}(y - f(x)) = (Df(x) + r(z - x))^{-1} - Df(x)^{-1}$$

By an analytic estimate (which we shall not prove here), we obtain the following:

$$\|\hat{r}(y - f(x))\| \leq 2\|Df(x)^{-1}\|^2\|r(z - x)\| \rightarrow 0$$

Using a similar estimate, we get continuity of the derivative of the local inverse:

$$\begin{aligned} \|Df^{-1}(y) - Df^{-1}(f(x_0))\| &= \|Df(f^{-1}(y))^{-1} - Df(x_0)^{-1}\| \\ &\leq 2\|Df(x_0)^{-1}\| \cdot \|Df(f^{-1}(y)) - Df(x_0)\| < 2\|Df(x_0)^{-1}\|\epsilon \end{aligned}$$

so using the same logic as differentiability, for all nets $y \rightarrow f(x_0)$ by continuity of f^{-1} , $f^{-1}(y) \rightarrow x_0$. By the continuity of Df , we have the continuity of Df^{-1} at $f(x_0)$. $\square$

To put this theorem more succinctly, if $f \in C^1$ at some point $x_0 \in \text{Dom}(f)$, then there exists $\delta > 0$ (provided by continuity of $Df(x)$ at x_0), where f has a local inverse in $f(B(x_0, \delta))$ such that $f^{-1} \in C^1$ at $f(x_0)$.

Lemma 1.9. *Given some matrix $A \in \mathcal{M}_{n \times k}(\mathbb{R})$ and an arbitrary submatrix $B \in \mathbb{M}_{m \times j}(\mathbb{R})$, we have that $\|B\| \leq \|A\|$.*

Proof. Consider the submatrix C , for which each column of C shares the same j column numbers as B with no row deletion. Then for an arbitrary $v \in \mathbb{R}^j$ where Bv is valid, we also have that Cv is valid.

$$\|Bv\| = \sqrt{(B_1 \cdot v)^2 + \dots + (B_m \cdot v)^2} \leq \|Cv\|$$

where B_i is the i -th row of B . The inequality follows from the fact that each column of B is a trimmed version of its respective counterpart in C , thus B can be obtained by row deletion of C . Now, we can add zeros in $v \in \mathbb{R}^j$ to get $v' \in \mathbb{R}^k$, so that $\|Cv\| = \|Av'\|$, thus:

$$\frac{\|Bv\|}{\|v\|} \leq \frac{\|Cv\|}{\|v\|} \leq \|A\|$$

Since $v \in \mathbb{R}^j$ was arbitrary, we have that $\|B\| \leq \|A\|$. $\square$

We now view the implicit function theorem which provides us the existence for a parametrization of a function given certain conditions at a certain point.

Theorem 1.5 (Implicit Function Theorem). *Consider $F : \mathbb{R}^{n+m} \rightarrow \mathbb{R}^m$ and $x \in \mathbb{R}^n$ and $y \in \mathbb{R}^m$. Then take the point $(x_0, y_0) \in \text{int}(\text{Dom}(F))$, such that:*

$$F(x_0, y_0) = 0$$

If we are given some $B((x_0, y_0), \delta_0)$ on which DF is continuous at the center and $D_y F$ is invertible at that point, then there exists some $\delta' > 0$ such that:

$$\forall x \in B(x_0, \delta') : y(x_0) = y_0 \wedge F(x, y(x)) = 0$$

where $y(x)$ is a unique parametrization on the ball $B(x_0, \delta')$ and is differentiable on the ball and C^1 at x_0 . Additionally:

$$\forall x \in B(x_0, \delta') : Dy(x) = -D_y F(x, y(x))^{-1} D_x F(x, y(x))$$

Proof. Consider the map $f : \mathbb{R}^{n+m} \rightarrow \mathbb{R}^{n+m}$ defined as the following:

$$f \left(\begin{bmatrix} x \\ y \end{bmatrix} \right) := \begin{bmatrix} T(x, y) \\ F(x, y) \end{bmatrix}$$

where $T(x, y) = x$. Then, we have that $Df(x, y) \in \mathcal{M}_{(n+m) \times (n+m)}(\mathbb{R})$:

$$Df(x, y) = \begin{bmatrix} D_x T(x, y) & D_y T(x, y) \\ D_x F(x, y) & D_y F(x, y) \end{bmatrix} = \begin{bmatrix} I & 0 \\ D_x F(x, y) & D_y F(x, y) \end{bmatrix}$$

We now observe that the continuity of $Df(x, y)$ at (x_0, y_0) follows from that of $DF(x, y)$ via the argument in Lemma 1.9 and the fact that the top two blocks are 0 matrices:

$$\begin{aligned} \|Df(x, y) - Df(x_0, y_0)\| &\leq \| [D_x F(x, y) - D_x F(x_0, y_0) \quad D_y F(x, y) - D_y F(x_0, y_0)] \| \\ &= \|DF(x, y) - DF(x_0, y_0)\| < \epsilon \end{aligned}$$

so the same δ_0 suffices (using the conclusion Lemma 1.9, we actually get equality). We now have a map, f , between same dimensional euclidean spaces, which is differentiable on a δ_0 -ball (follows from the existence of DF) and continuous at the center. We now need to show that $Df(x_0, y_0)$ is invertible to invoke the Inverse Function Theorem.

We claim that A is the inverse of $Df(x_0, y_0)$ such that:

$$A \circ Df(x_0, y_0) = Df(x_0, y_0) \circ A = I$$

examining the structure of $Df(x, y)$, we have that $DT(x, y) \in \mathcal{M}_{n \times (n+m)}(\mathbb{R})$, so since $x \in \mathbb{R}^n$ (which means it is a vector $(x_1, \dots, x_n)$), we have that $I \in \mathcal{M}_{n \times n}(\mathbb{R})$ and $0 \in \mathcal{M}_{m \times n}(\mathbb{R})$. Similarly $D_x F(x_0, y_0) \in \mathcal{M}_{m \times n}(\mathbb{R})$ and $D_y F(x_0, y_0) \in \mathcal{M}_{m \times m}(\mathbb{R})$.

In order to maintain the same dimensions and for block matrix multiplication to work:

$$A := \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$$

we claim that each block has the same dimension as its respective counterpart in $Df(x_0, y_0)$. Then we want the following:

$$\begin{bmatrix} A_{11}I + A_{12}D_x F(x_0, y_0) & A_{12}D_y F(x_0, y_0) \\ A_{21}I + A_{22}D_x F(x_0, y_0) & A_{22}D_y F(x_0, y_0) \end{bmatrix} = \begin{bmatrix} I & 0 \\ 0 & I \end{bmatrix}$$

In order to ensure the top row evaluates to the desired result, we can set $A_{11} = I, A_{12} = 0$. Then since $D_y F(x_0, y_0)$ is invertible, we can set $A_{22} = D_y F(x_0, y_0)^{-1}$, which means $A_{21} = -D_y F(x_0, y_0)^{-1} D_x F(x_0, y_0)$. So:

$$A = \begin{bmatrix} I & 0 \\ -D_y F(x_0, y_0)^{-1} D_x F(x_0, y_0) & D_y F(x_0, y_0)^{-1} \end{bmatrix}$$

We can check that this form of A serves as the correct right inverse. Now, we have that f is differentiable on a δ_0 ball with an invertible and continuous derivative at (x_0, y_0) , the center of the ball. We can apply Theorem 1.4 now, which gives us some local inverse in $B((x_0, y_0), \delta) \subseteq B((x_0, y_0), \delta_0)$, such that:

$$\forall (x, y) \in B((x_0, y_0), \delta) : f^{-1}(f(x, y)) = f^{-1} \left(\begin{bmatrix} x \\ F(x, y) \end{bmatrix} \right) = \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\forall (u, v) \in f(B((x_0, y_0), \delta)) : f(f^{-1}(u, v)) = \begin{bmatrix} u \\ v \end{bmatrix}$$

By the first relation of f^{-1} being a left inverse, let us say that:

$$f^{-1} \left(\begin{bmatrix} x \\ F(x, y) \end{bmatrix} \right) = \begin{bmatrix} h_1(x, F(x, y)) \\ h_2(x, F(x, y)) \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}$$

Then by the second relation of right inverse:

$$f \left(\begin{bmatrix} h_1(u, v) \\ h_2(u, v) \end{bmatrix} \right) = \begin{bmatrix} h_1(u, v) \\ F(h_1(u, v), h_2(u, v)) \end{bmatrix} = \begin{bmatrix} u \\ v \end{bmatrix}$$

which gives us that $F(h_1(u, v), h_2(u, v)) = v$ and $h_2(x, F(x, y)) = y$. Thus $h_2(x_0, 0) = y_0$ and $F(x_0, h_2(x_0, 0)) = 0$. Now observe that $f(x_0, y_0) = (x_0, 0)$ so $(x_0, 0) \in f(B((x_0, y_0), \delta))$.

Since f is an open map (as f^{-1} is continuous on the image of the open ball), then there exists some $\delta' > 0$ such that $B((x_0, 0), \delta') \subseteq f(B((x_0, y_0), \delta))$. Additionally $B(x_0, \delta') \times \{0\} \subseteq B((x_0, 0), \delta')$. So, we can define $y(x) := h_2(x, 0)$ such that:

$$\forall x \in B(x_0, \delta') : F(x, y(x)) = F(x, h_2(x, 0)) = 0 \wedge F(x_0, y(x_0)) = F(x_0, y_0) = 0$$

Because Df^{-1} exists, we can say that Dy exists as $D_x h_2(x, 0)$, and is continuous by the continuity of Df^{-1} at $f(x_0, y_0) = (x_0, 0)$ and Lemma 1.9. We can also prove its uniqueness, but we shall not do that here. Observe that we can also have the level set constraint $F(x, y) = c$ for some $c \in \mathbb{R}$, and the proof would be the same. By chain rule, we have:

$$\forall x \in B(x_0, \delta') : (F \circ G)(x) = c$$

where $G(x) = (x, y(x))$ such that $G : \mathbb{R}^n \rightarrow \mathbb{R}^{n+m}$. Then:

$$DF(x, y(x)) \circ DG(x) = \begin{bmatrix} D_x F(x, y(x)) & D_y F(x, y(x)) \end{bmatrix} \begin{bmatrix} I \\ Dy(x) \end{bmatrix} = 0$$

$$D_x F(x, y(x)) + D_y F(x, y(x)) Dy(x) = 0 \implies Dy(x) = -D_y F(x, y(x))^{-1} D_x F(x, y(x))$$

We have the invertibility of $D_y F(x, y)$ for $(x, y) \in B((x_0, y_0), \delta)$, since $Df^{-1}(x, y) = Df(x, y)^{-1}$ from the implicit function theorem. Let us call this inverse A ; then by the previous result, we know that A_{22} serves as a left inverse:

$$\begin{aligned} & \begin{bmatrix} I & 0 \\ D_x F(x, y) & D_y F(x, y) \end{bmatrix} \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \\ &= \begin{bmatrix} A_{11} & A_{12} \\ D_x F(x, y) A_{11} + D_y F(x, y) A_{21} & D_x F(x, y) A_{12} + D_y F(x, y) A_{22} \end{bmatrix} = \begin{bmatrix} I & 0 \\ 0 & I \end{bmatrix} \end{aligned}$$

This forces $A_{11} = I$ and $A_{12} = 0$, thus the bottom right simplifies to:

$$D_y F(x, y) A_{22} = I$$

showing that A_{22} also serves as a right inverse, thereby proving its invertibility. It follows by subset inclusion that this holds for all $x \in B(x_0, \delta')$. $\square$

1.3 Configuration Spaces

The configuration space of a robot is the space of all various positions the parts of a robot or the robot itself can be in and is topologically a manifold (e.g. torus, cylinder, etc.). The state space is an extension of this.

We will denote the configuration space as $\mathcal{C}$, and the dimension of the configuration space is called the degrees of freedom. Take for example the stationary arm robot with 4 revolute joints and links, we define:

$$\mathcal{C} := \{[\theta_1, \theta_2, \theta_3, \theta_4]^T : (\forall i \in \{1, 2, 3, 4\} : \theta_i \in [0, 2\pi))\}$$

We call $(\theta_1, \theta_2, \theta_3, \theta_4)$ as coordinates, which to be more precise, are component functions of a local chart:

$$\theta : U \rightarrow \mathbb{R}^4$$

where $U \subseteq \mathcal{C}$ and U is open. Thus:

$$\forall p \in U : \theta(p) = \begin{bmatrix} \theta_1(p) \\ \theta_2(p) \\ \theta_3(p) \\ \theta_4(p) \end{bmatrix} \in \mathbb{R}^4$$

The configuration space of a Cartesian robot (also called gantry or rectilinear) is $\mathbb{R}^n$ depending on the dimensions (typically $\mathbb{R}, \mathbb{R}^2, \mathbb{R}^3$). A good example is a 3D printer, which has a nozzle that can move in the x, y directions and up/down, which constitutes the z direction.

1.3.1 Holonomic Constraints and Implicit Functions

Definition 1.4 (Holonomic Constraints). *Given $\dim(\mathcal{C}) = n$, a holonomic constraint is one where for some $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$ such that for some $k \in \mathbb{R}^m$:*

$$F(c_1, \dots, c_n) = k$$

where $(c_1, \dots, c_n)$ are the coordinates of $\mathcal{C}$ and $n \geq m$.

Note: Going forward, we shall assume that holonomic constraints are C^1 on their domain, though they need not be.

In this definition, we treat the coordinates as Euclidean vectors in $\mathbb{R}^n$. In reality, the map being used is $F : \mathcal{C} \rightarrow \mathbb{R}^m$, though the results we shall further introduce holds for both cases of F (and can be further extended to when F is a C^1 map between manifolds).

Lemma 1.10. *A linear transformation $T : V \rightarrow W$ is surjective if and only if its matrix representation has full row rank.*

Proof. This follows from the rank nullity theorem, for which the rank should be $\dim(W)$. If we select bases β_V, β_W , then $[T]_{\beta_V}^{\beta_W}$ has $\dim(W)$ rows. $\square$

We now define the following notion:

Definition 1.5 (Regular Level Set). *The following is the definition of a level set for some $c \in \mathbb{R}^m$ given $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$:*

$$F^{-1}(k) := \{c \in \mathbb{R}^n : F(c) = k\}$$

A regular level set is one whose total derivative (aka differential) is surjective (or equivalently full row rank) at all $c \in F^{-1}(k)$.

In the case of our "approximate" map $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$, our total derivative is the Jacobian $DF(c) \in \mathcal{M}_{m \times n}(\mathbb{R})$. This is actually equivalent to one of the sufficient conditions in Implicit Function Theorem for $D_y F(x_0, y_0)$ to be invertible.

Since $n \geq m$, let $n = m + j$. We know that $DF(c)$ has m rows, which means it has rank m (given $F^{-1}(k)$ is a regular level set and $c \in F^{-1}(k)$). Then take the m linearly independent columns (as row rank is equal to column rank) and label the coordinate directions of their partials as $e_{n-m+1}, \dots, e_n$.

Then label the remaining directions for the j linearly dependent columns as $e_1, \dots, e_{n-m}$. Then:

$$DF(c) = \begin{bmatrix} D_x F(c) & D_y F(c) \end{bmatrix}$$

Since $D_y F(c)$ is a square matrix of full rank, it is invertible. The converse automatically follows. Additionally, we can write $F : \mathbb{R}^{m+j} \rightarrow \mathbb{R}^m$, where $x \in \mathbb{R}^j, y \in \mathbb{R}^m$.

Definition 1.6 (Spivak). *A subset M of $\mathbb{R}^n$ is an $n - m$ dimensional manifold if for all $x \in M$:*

1. *There exists some open neighborhood of U of x and some $V \subseteq \mathbb{R}^n$*
2. *There exists a diffeomorphism $h : U \rightarrow V$ such that:*

$$h(U \cap M) = V \cap (\mathbb{R}^{n-m} \times \{0\})$$

where a diffeomorphism is a C^k bijection with a C^k inverse. When we set $k = 0$, we obtain a home-

omorphism, or a bijection between two topological spaces which is continuous and has a continuous inverse.

Lemma 1.11. *Given a holonomic constraint as per Definition 1.4, the regular level set $F^{-1}(k)$ is the constrained configuration space C' with dimension $n - m$.*

Proof. In the proof for the Implicit Function Theorem, we obtained a function $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$, where:

$$f(x, y) := \begin{bmatrix} x \\ F(x, y) \end{bmatrix}$$

given $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$. The inverse function theorem gave us that f has a local inverse in some ball $B((x_0, y_0), \delta)$ which is differentiable in the image and continuous at $f(x_0, y_0) = (x_0, k)$.

In this case, let (x_0, y_0) be an arbitrary point in $F^{-1}(k)$. Then let $U = B((x_0, y_0), \delta)$, $V = f(B((x_0, y_0), \delta))$ and f be the diffeomorphism in question, then:

$$\forall (x, y) \in U \cap M : f(U \cap M) = (x, k) \in f(B((x_0, y_0), \delta)) \cap (\mathbb{R}^{n-m} \times \{k\})$$

Now consider some $(x, k) \in f(B((x_0, y_0), \delta)) \cap (\mathbb{R}^{n-m} \times \{k\})$, we want to show that $(x, k) \in f(U \cap M)$. We have the identity:

$$\forall (u, v) \in f(B((x_0, y_0), \delta)) : F(u, h_2(u, v)) = (F \circ f^{-1})(u, v) = v$$

Thus $F(f^{-1}(x, k)) = k$, which tells us that $f^{-1}(x, k) \in M$ and:

$$(x, k) \in f(B((x_0, y_0), \delta)) \cap (\mathbb{R}^{n-m} \times \{k\}) \implies f^{-1}(x, k) \in B((x_0, y_0), \delta)$$

So $f^{-1}(x, k) \in U \cap M$, which gives us that $(x, k) \in f(U \cap M)$. □

Lemma 1.11 also gives us an important consequence (via conclusion of the implicit function theorem), that is, the number of variables required to represent the variables/behavior of a system can be locally reduced around a point in the level set by implicit parametrization.

1.4 More on Manifolds*

An interesting result is that the inverse and implicit function theorems are logically equivalent. The provided proofs already show that given the inverse function theorem, we can obtain the implicit function theorem.

For the converse, assume we have implicit function theorem. Then consider the function $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ and:

$$F(x, y) = f(x) - y = 0$$

Then the graph of f is the level set $F^{-1}(0)$. Assume we are also given the sufficient conditions required by the inverse function theorem, that is:

1. $f(x_0) = y_0$ and $Df(x_0)$ is invertible
2. $Df(x)$ exists on an open ball around x_0 and is continuous at x_0

Then:

$$DF(x, y) = [Df(x) \quad -I]$$

$$\|DF(x, y) - DF(x_0, y_0)\| \leq \|Df(x) - Df(x_0)\| < \epsilon$$

Thus $F \in C^1$ at (x_0, y_0) by the continuity of Df , so the same δ suffices. Since the value of y does not matter, we can simply consider the neighborhood $B((x_0, y_0), \delta)$. Now because $Df(x_0)$ is invertible and continuous at

x_0 , if we take $\epsilon = (1/2)\|Df(x_0)\|^{-1}$, then via the alternate proof of part one in the inverse function theorem, we get that f is also injective on $B(x_0, \delta)$.

Furthermore, we have that $D_x F(x_0) = Df(x_0)$ is invertible and given the continuity of DF , we apply implicit function theorem to see that:

$$\exists \delta' > 0, \forall y \in B(y_0, \delta') : F(g(y), y) = f(g(y)) - y = 0$$

where g is differentiable in the ball and Dg continuous at y_0 . So g is a right inverse. Similarly, given some x such that $x = g(y)$ for some $y \in B(y_0, \delta')$ implies that $x = g(f(x))$ showing that it is a left inverse for $x \in g(B(y_0, \delta'))$. However, we cannot necessarily say that $f : g(B(y_0, \delta')) \rightarrow B(y_0, \delta')$ is a bijection (I believe we can actually prove it is by showing g is an injection, but that is still not enough to obtain a diffeomorphism. The remaining proof will remain the same).

Consider the set $f^{-1}(B(y_0, \delta'))$, which is open due to the continuity of f . We have that $x_0 \in f^{-1}(B(y_0, \delta'))$, so we consider $B(x_0, \delta_m)$, where:

$$\delta_m := \min\{\delta, \delta''\}$$

where δ is from the injectivity and δ'' is from the interior point property. This means $f : B(x_0, \delta_m) \rightarrow f(B(x_0, \delta_m))$ is injective, surjective, differentiable within the ball, and Df continuous at x_0 . Additionally:

$$B(x_0, \delta_m) \subseteq f^{-1}(B(y_0, \delta')) \implies f(B(x_0, \delta_m)) \subseteq B(y_0, \delta')$$

Thus for all $y \in f(B(x_0, \delta_m))$, $F(g(y), y) = 0$, so g is a left and right inverse as per our previous argument, which gives us $g(f(B(x_0, \delta_m))) = B(x_0, \delta_m)$. Additionally, g is differentiable on this image ball and Dg continuous at y_0 by subset inclusion.

Therefore $f : B(x_0, \delta_m) \rightarrow f(B(x_0, \delta_m))$ is the diffeomorphism in question, with a local continuously differentiable (at y_0) inverse $g : f(B(x_0, \delta_m)) \rightarrow B(x_0, \delta_m)$.

1.5 Pfaffian Constraints*

This is currently beyond my understanding... The topic requires integration on manifolds.

1.6 Examples in Robotics

2 Constrained State Spaces